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Contents

Contact 1
Personal Details 1
Current Research 1
Skill Profile 2
Education Experiences 2
Seclected Project Experiences 3
Publications 3
Patents 4
Honors and Awards 4

✧ Contact

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✧ Personal Details

Given/Family Name: Chuang Yang | **Gender**: Male | **Birth**: 1992 | **Marital status**: Single
Current status: 3rd-year PhD Student at NUAA (5-year full-time doctor's degree program).
Language: Mandarin Chinese, English (full professional proficiency)
Personal Homepage: www.yangchuang.link

✧ Current Research

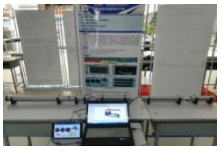
- **Major**: Guidance, Navigation and Control
- **Research interests**
  Brain-inspired Swarm Navigation in Robotics and Neuroscience, Integrated Navigation System (INS/GNSS/Magnetic), Bio-inspired Self-Organizing Robotic System, Neuromorphic Swarm.
- **PhD thesis topic:** Brain-inspired swarm navigation for UAVs
  Abstract: This research project (www.neuromorphicswarm.com) will fuse neural basis of bat's 3D navigation & swarm navigation, collective behavior of animal groups and the decentralized cooperative

navigation by employing neuroscience-oriented artificial intelligence method (e.g. CANN, ML, SNN) to develop a new class of brain-inspired swarm navigation architecture—towards Neuromorphic Swarm—of UAV Swarm.

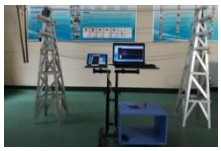
- **Professional association**
  Member of Royal Institute of Navigation, Member of Chinese Society of Aeronautics and Astronautics, Member of Chinese Association for Artificial Intelligence, Member of Chinese Association of Automation.

✧ Skill Profile

- **Programming**
  MATLAB, C, LabVIEW, Python
- **Web Design**
  Hugo, Github
- **Open Source Program**
  RatSLAM: a brain-based SLAM system
  NeuroSLAM: a brain-inspired SLAM system for 3D environments
  SwarmLab: an open source MATLAB Drone Swarm Simulator
  GNSS/INS: an Integrated Navigation System Simulator (designed by my group).
- **Embedded Hardware**
  Arduino, STM32
- **Design of Electronic System**
  (*I became interested in electronic products in childhood, greatly influenced by my parents who were all maintanance workers for electronic products, such as television, amplifier, etc.*)
  Two seclected designs of electronic system:



IMU/Magnetic sensors-based Integrated Navigation System of Measurement While Drilling
Third Prize in the 4th Petroleum Equipment Innovative Design Competition for China postgraduate (**team leader**), 09/2017.



IMU-based On-line Monitoring System of Transmission Line Galloping
Second Prize in the 12th Electronic Design Competition for China postgraduate (**team leader**), 07/2017.

✧ Education Experiences

- **04/2018—present**
  Nanjing University of Aeronautics and Astronautics, China
  PhD student of Navigation, Guidance and Control
  Supervisor: Prof. Jianye Liu
  Co-supervisor: Prof. Zhi Xiong
  Thesis Topic: Brain-inspired swarm navigation of UAVs
- **07/2020—07/2020**
  Certificate of Completion of Neuromatch Academy (an online school for Computational Neuroscience)
  Interactive student of Computational Neuroscience
  Mentor: Dr. Torres Gelsy (Associate Professor, Universtity of Pittsburgh)
  Course Project: How does task related activity interact with intrinsic activity of the brain?
- **09/2015—12/2017**

Henan Polytechnic University, China

Master of Navigation, Guidance and Control (***I received my M.S. degree ahead of schedule by six months due to the excellent academic performance***)

Supervisor: Dr. Jinxian Yang

Final thesis: Research on INS/magnetic integrated navigation technique for measurement while drilling

- **09/2011—07/2015**

  Henan Institute of Science and Technology, China

  Bachelor of Electrical Engineering and Automation (GPA: 3.68/4.0)

✧ Seclected Project Experiences

- **Brain-inspired navigation & swarm navigation**

1. China, National Natural Science Foundation of China, Grant No. 61873125. **Brain-inspired intelligent cooperative navigation for dense UAV swarm,** 01/2019-12/2022. (main participant, my Ph.D. thesis topic).
2. China, Shanghai Aerospace Science and Technology Innovation Fund, Grant No. SAST2019-085. **Brain-inspired positioning technology for intelligent aircraft**, 12/2019-12/2021. (main participant).

- **Integrated navigation**

3. China, National Natural Science Foundation of China, Grant No. U1404510. **Key technologies of MEMS inertial navigation for measurement while drilling**, 01/2015-12/2017. (main participant)
4. China, Key Science and Technology Program of Henan Province, Grant No. 172102210289. **Research on on-line monitoring system of transmission line galloping based on inertial technology**, 09/2016-09/2017. (main participant)

✧ Publications

- **Brain-inspired navigation & swarm navigation**

1. **Chuang Yang**, Zhi Xiong, Jianye Liu, Lijun Chao, Yudi Chen. **Brain-inspired metric navigation of UAV based on model of multi-scale grid cells**[J]. IEEE Transactions on Cognitive and Developmental Systems (Under review, 2020.08)
2. **Chuang Yang**, Jianye Liu, Zhi Xiong, Lijun Chao. **Brain-inspired vector navigation method for UAV based on model of multi-scale grid cells**[J]. Journal of Chinese Inertial Technology, 2020, 28(02):179-185. (EI, in Chinese)
3. **Chuang Yang**, Jianye Liu, Zhi Xiong. **A new paradigm of brain-inspired navigation integrating perception and decision**[EB/OL]. article.xuexi.cn, 2020.03.04. (in Chinese)
4. **Chuang Yang**, Jianye Liu, Zhi Xiong, Jun Xiong. **Brain-inspired navigation technology integrating perception and action decision: A review and outlook**[J]. Acta Aeronautica et Astronautica Sinica, 2020, 41(1):323280. doi:10.7527/S1000-6893.2019.23280. (EI, in Chinese)
5. Jianye Liu, **Chuang Yang**, Zhi Xiong, Jizhou Lai, Jun Xiong. **Attractor neural network-based brain-inspired navigation technology for UAV**[J]. Navigation Positioning & Timing, 2019, 6(5): 52-60. (in Chinese)
6. **Chuang Yang**, Jianye Liu. **Brain-inspired navigation technology and its applications in unmanned autonomous system**[J]. Naas & Inertal Technology, 2019, 200+201:151-152. (in Chinese)

- **Integrated navigation**

7. Jinxian Yang, **Chuang Yang**, Shuanglei Li, et al. **Vibration analysis of transmission lines based on MIMU**[J]. Journal of Vibration and Shock, 2018, 37(11): 230-236. (EI, in Chinese)
8. Jinxian Yang, **Chuang Yang**, Jinbao Jiang. **A robust mag/INS-based orientation estimation algorithm for measurement while drilling**[J]. IEEE Sensors Journal, 2017, 17(23): 7971–7980. (SCI)
9. Jinxian Yang, **Chuang Yang**. **MEMS gyro signal reconstruction via layerwise and adaptive MP**[J].

Optics and Precision Engineering, 2017, 25(12): 3160-3168. (EI, in Chinese)

10. Jinxian Yang, **Chuang Yang**, Zhitao Jiang. **Adaptive extraction of gravity information for drilling tool with magnetic and inertial integrated system**[J]. Journal of china coal society, 2017, 42(12): 3331-3337. (EI, in Chinese)
11. Jinxian Yang, **Chuang Yang**. **Signal extraction of mems gyro utilized in MWD based on sparse representation**[J]. Chinese Journal of Sensors & Actuators, 2017, 30(11): 1677-1683. (in Chinese)
12. Jinxian Yang, **Chuang Yang**. **Gyro drift estimation and compensation based on gravity quaternion**[J]. Chinese Journal of Sensors and Actuators, 2017, 30(8): 1187-1192. (in Chinese)

✧ Patents

1. **Chuang Yang**, Zhi Xiong, Jianye Liu, etc. **A brain-inspired navigation method in 3D environment based on path integration of multi scale grid cells**[P]. China Invention Patent, (2020-08, Under review)

✧ Honors and Awards

10/2017: National Scholarship for Postgraduate

09/2017: Third Prize in the 4th Petroleum Equipment Innovative Design Competition for China postgraduate, ***INS/Magnetic-based integrated navigation system of measurement while drilling***. (**team leader**)

07/2017: Second Prize in the 12th Electronic Design Competition for China postgraduate, ***INS-based on-line monitoring system of transmission line galloping***. (**team leader**)

09/2015-01/2018: Full scholarship for Postgraduate

11/2013: National Scholarship for Encouragement

03/2013: Third-level Certificate for National Computer

11/2012: National Scholarship for Encouragement

09/2011-07/2015: The First Prize Scholarship for four years